

WIA1006 - CROPGPT

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THEME

CROP PRODUCTION

BY DISTRICT



THE PROBLEM

The variation in crop performance across different districts makes it difficult for farmers to choose the most productive crop to plant. Without reliable methods to predict and compare crop yields, decisions are often based on guesswork, leading to lower agricultural productivity and inefficient land use.

Crop productivity varies across regions, different crop grow better in different environments.

- How can farmers know which *crop species* will give them *the best yield* in their specific district?
- What *factors* influence the *productivity* of different crops in various regions?
- Is it possible to *use past data* to predict which crops will perform better in certain districts?
- Can *temperature, humidity, and planted area* data be used to guide crop selection more effectively?



OUR MAIN FOCUS

Task:

To identify the most suitable crop species to plant in a specific district.

Performance:

By predicting the production of each crop species in each district, enabling comparison of productivity across different crops and regions.

Experience:

Using data from 2016 consisting of planted area, district, crop species, and temperature-humidity index (derived from temperature & humidity).

DATA COLLECTION:

Datasets used:

1. Crop Production by District (OpenDOSM)
2. Crop Area by District (OpenDOSM)
3. Climate by State (WorldData.info)

DATA PREPROCESSING

- Identify mismatched values
 1. **crop_type**: 'vegetables' in area dataset but 'vegetable' in production dataset.
 2. **crop_species**: 'angled_loofah' in area dataset but 'luffa' in production dataset
 3. **crop_species**: 'snake_gourd', 'mint', 'chinese_potato' exist ONLY in production dataset
- Dropped mismatched values and standardise inconsistencies
 1. **Replaced** 'vegetables' with 'vegetable'.
 2. **Replaced** 'luffa' with 'angled_loofah'.
 3. **Dropped** 'snake_gourd', 'mint', and 'chinese_potato'

DATA PREPROCESSING

- Merge datasets
 - Merged on exact matched '**date**', '**state**', '**district**', '**crop_type**', and '**crop_species**'
- Drop missing or invalid rows
 - Drop **missing** or **zero values** in 'planted area' and 'production' using dropna
- Drop indicator column
 - Drop '**_merge**' column

FEATURE SELECTION

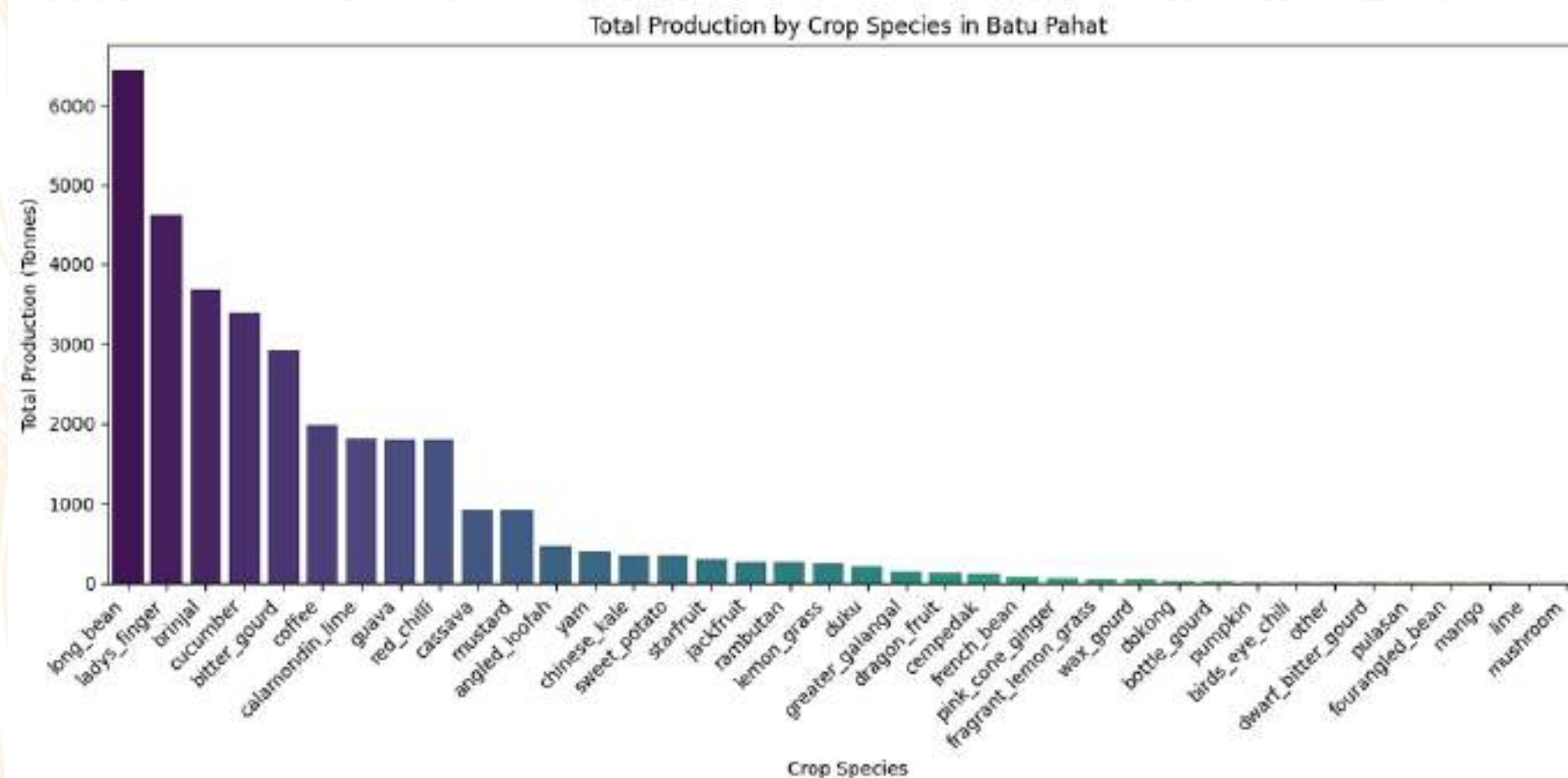
- Feature engineering
 - Temperature-humidity index (THI) is computed to combine average temperature and humidity into a single feature that represents the climate stress level of the crops.
$$\text{THI} = \text{Avg_Temp} - (0.55 - 0.0055) * \text{Humidity} * (\text{Avg_Temp} - 14.5)$$
- Features chosen:
 1. Target feature: **'production'**
 2. Selected features: **'planted_area', 'crop_species', 'district', 'THI'**
- Features excluded:
 - 'date'** redundant; **'state'** redundant; **'rainfall'** low importance score; **'crop_type'** redundant as we are using 'crop_species'

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HOW CAN FARMERS KNOW WHICH CROP SPECIES WILL GIVE THEM THE BEST YIELD IN THEIR SPECIFIC DISTRICT?

Based on the initial visualised graph, we identified that crop yields vary significantly by region due to environmental and spatial factors.

For example, in district **Batu Pahat**, **long bean** (*kacang panjang*) crops consistently produced the highest average yield of **63000 tons/hectare**.



We believe these comparisons & yield performance could be modeled by machine learning models, enabling farmers to predict the most suitable crop types for each district.

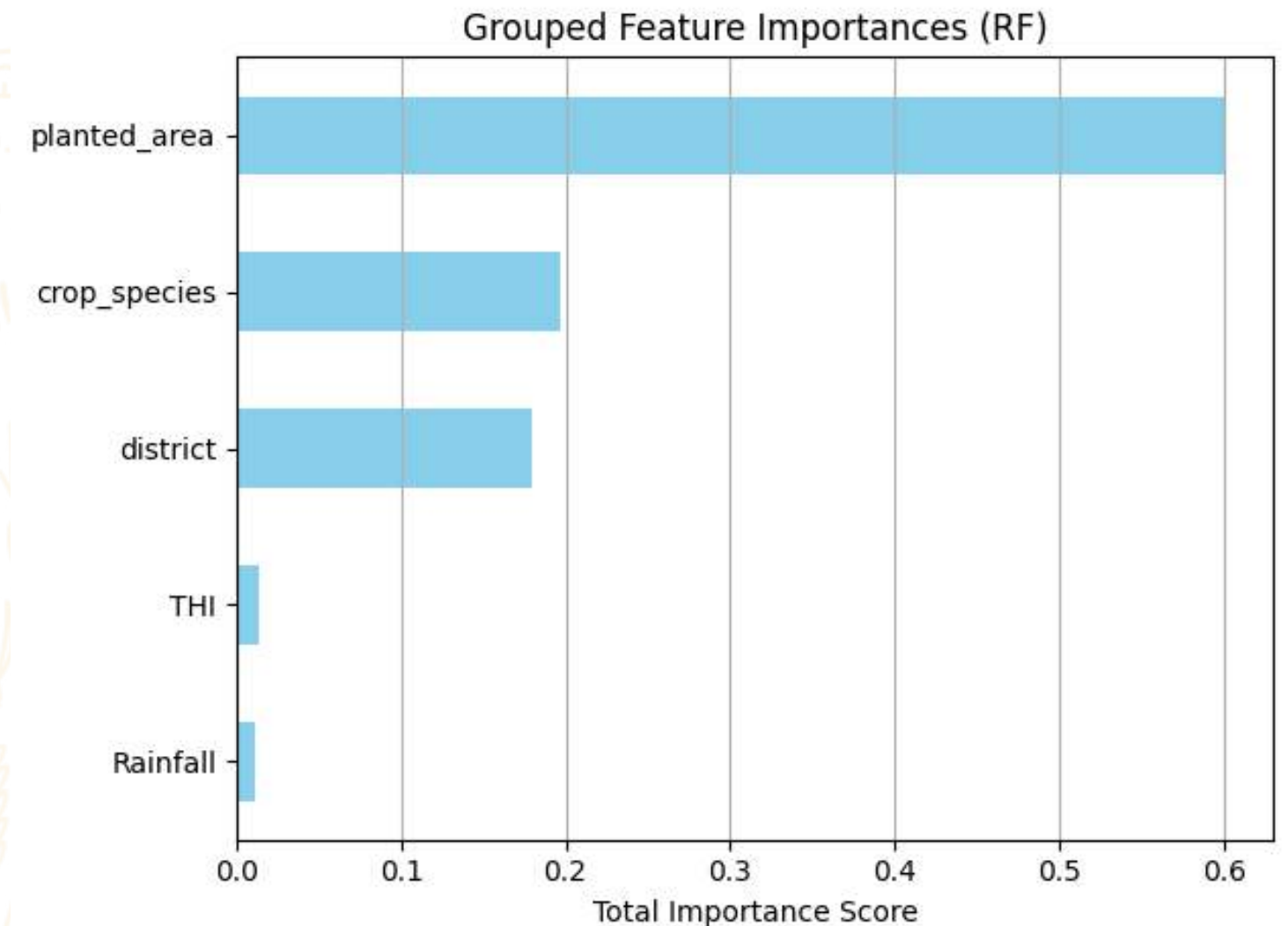


Kacang panjang thrives in warm environment & have a short life cycle; which explains the high yield

WHAT FACTORS INFLUENCE THE PRODUCTIVITY OF DIFFERENT CROPS IN VARIOUS REGIONS?

From our feature importance analysis, the most influential factors for crop productivity were:

- **Planted Area:** The more land dedicated to a crop, the higher the yield tends to be, likely due to better focus and farming resources.
- **Crop Species:** Some crops naturally perform better than others due to the local conditions and high humidity in Malaysia.
- **District:** Different districts have different climates, soil, and farming practices that affect how well crops grow.
- **Temperature-Humidity Index (THI):** Crops grow best in areas where the heat and humidity are within a comfortable range for that plant.

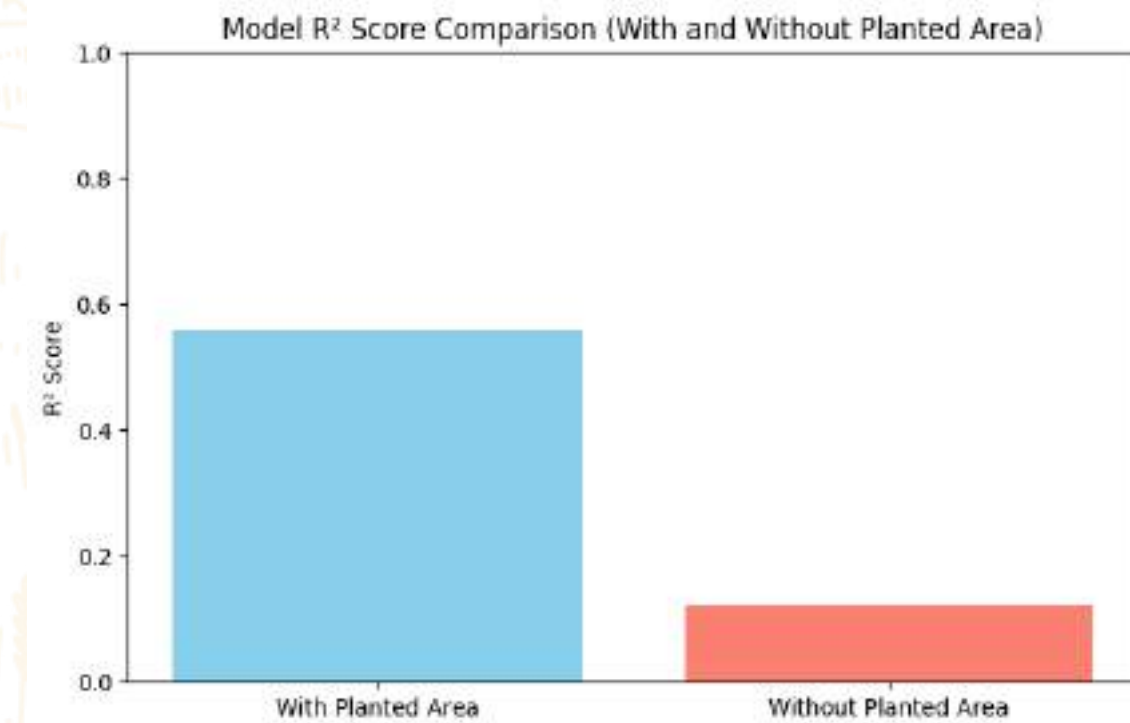


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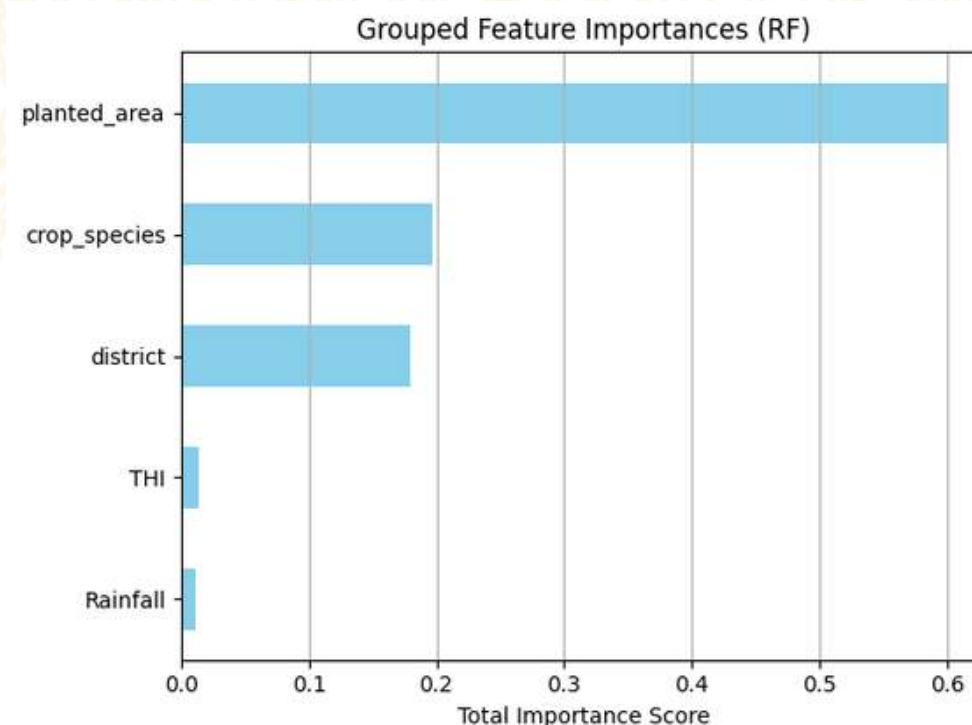
IS IT POSSIBLE TO USE PAST DATA TO PREDICT WHICH CROPS WILL PERFORM BETTER IN CERTAIN DISTRICTS?

Yes. We can use any suitable model on historical data from past years to **predict suitable crop types for each district**. Ideally, the model should be able to achieve accuracy on test set.

However, we found that the model may **heavily relied on planted area**, which risks reinforcing past decisions rather than identifying optimal crop choices.



R² value decreased when planted area is removed



Planted area is the most important feature, so the model might just be learning: "If a large area was used for Crop A, then Crop A is probably the best to plant again."

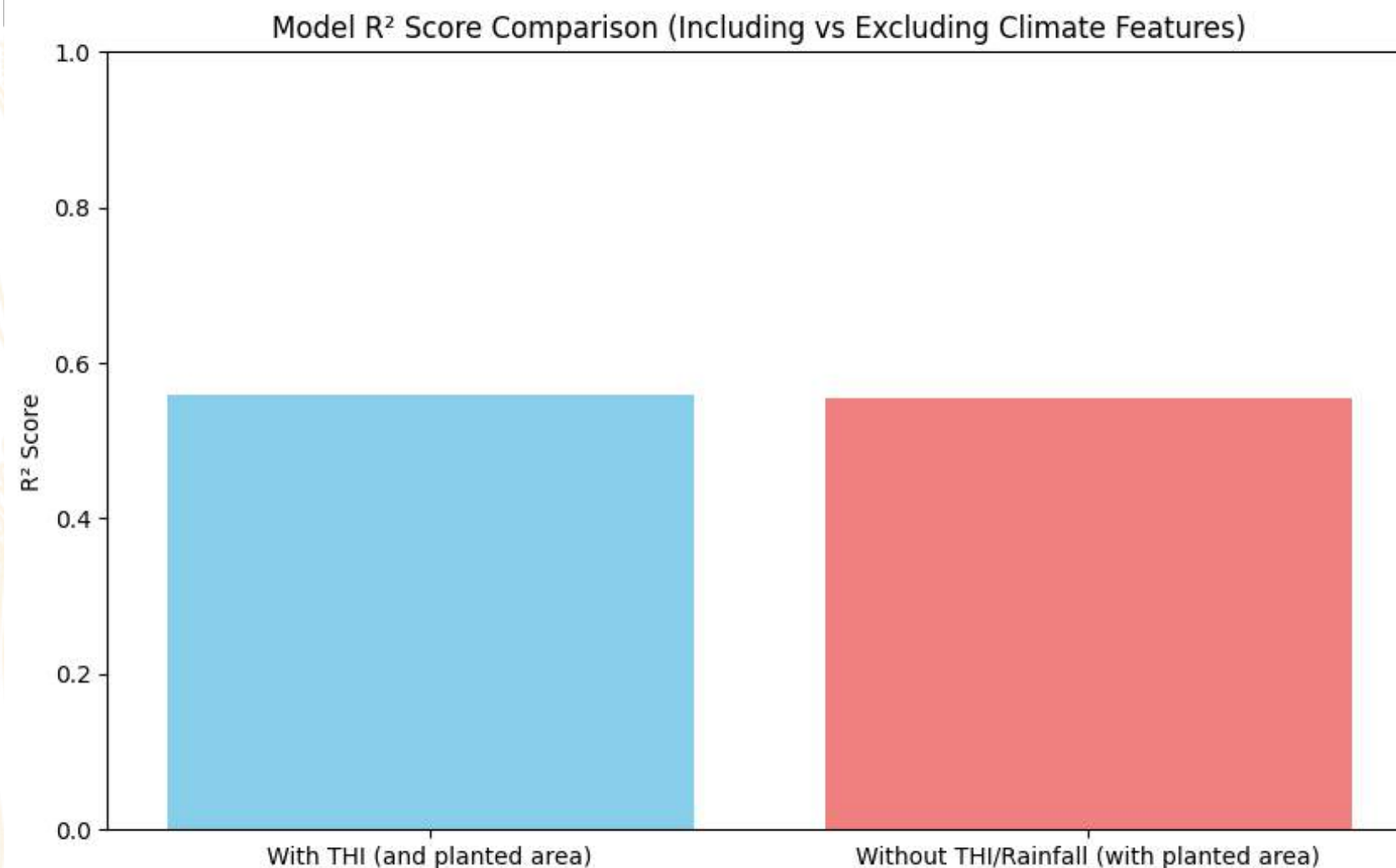
The model is probably **overfitting** to *past human choices*, and is not really predictive, and rather descriptive replication.

To avoid this, other environmental variables such as *temp*, *rainfall* and *humidity* should be incorporate more when training the model.

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CAN TEMPERATURE, HUMIDITY, AND PLANTED AREA DATA BE USED TO GUIDE CROP SELECTION MORE EFFECTIVELY?

Yes, **temperature and humidity** play a huge role in improving model predictions. When we included these environmental variables, our model's achieved a **slightly higher R2 value** (~0.55) .



This indicates that although planted area is a critical feature, including planted area alone *isn't enough* since it may reflect past human choices, and not optimal conditions. With THI, even a **small increase in R²** suggests the model is learning from **real agro-environmental conditions**, not just *past decisions* blindly.



Large planted areas may indicate high local expertise and resources for that crop.

MODEL SELECTION

Task Type: Supervised Regression

Target Variable: Production (tonnes)

Key Features:

- District
- Crop Species
- Planted Area
- Temperature-Humidity Index (THI)

EVALUATED MODELS

**Random Forest
Regressor**

**Gradient Boosting
Regressor**

**Linear
Regression**

**Support Vector
Regressor (SVR)**

**K-Nearest
Neighbours (KNN)**

MODEL TRAINING

Techniques applied:

One-Hot Encoding

handle_unknown = 'ignore' is used to ignore any district or crop species in the test set that are not present in the training set.

Scaling

StandardScaler is used to bring **planted_area** and **THI** to a similar scale so that the models focus on *how the numbers vary*, rather than their original values.

Log-Transform Target

log1p is used to stabilise target scale by *compressing large values and handling zero values* to make the data distribution less skewed.

Pipeline

To connect the data pre-processing steps and the models into a *single, consistent workflow*.

Data Split

80% Training
20% Testing

MODEL PERFORMANCE (WITHOUT TUNING)

	Model	R ²	Score	MAE	RMSE
3	Support Vector Regressor	0.6613	166.6812	536.1615	
0	Random Forest	0.5594	192.5693	611.4859	
4	K-Nearest Neighbors	0.4721	202.7463	669.3476	
1	Gradient Boosting	0.3599	224.2572	737.0540	
2	Linear Regression	0.2421	192.5150	802.0453	

HYPERPARAMETER TUNING

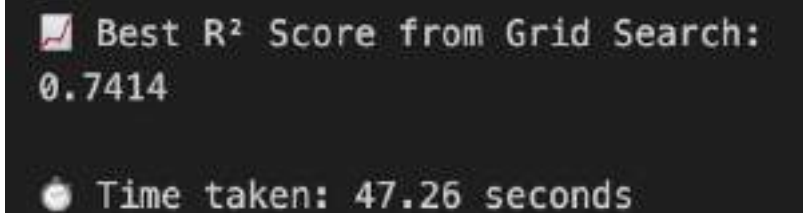
To improve performance of the Support Vector Regressor (SVR) model, GridSearchCV with 5-fold cross-validation is applied.

The best combination of key hyperparameters are:

Hyperparameter	Description
kernel	Type of kernel function ('rbf', 'poly')
C	Regularisation parameter ([1, 10, 100])
gamma	Kernel coefficient ('scale', 0.01, 0.1)
epsilon	Epsilon in the ϵ -insensitive loss function (0.1,0.2)

HYPERPARAMETER TUNING

- The pipeline combined one-hot encoding and scaling with the SVR model to ensure consistent transformation during training and testing.
- After training 36 different configurations with 180 fits, GridSearchCV selected the best parameters based on the R^2 score, which is **0.7414**.
- The optimised model was then evaluated on the unseen test set to assess real-world performance.
- Total tuning time on Colab: 285.69 seconds, or ~5 minutes (with 180 fits)
- Total tuning time when run locally: 47.26 seconds (with 180 fits)

A terminal window screenshot with a dark background and light-colored text. It displays the results of a Grid Search, showing the best R-squared score and the time taken for the search.

```
Best R2 Score from Grid Search:  
0.7414  
  
Time taken: 47.26 seconds
```


MODEL PERFORMANCE

(before tuning)

	Model	R^2	Score	MAE	RMSE
3	Support Vector Regressor	0.6613	166.6812	536.1615	
0	Random Forest	0.5594	192.5693	611.4859	
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(after tuning SVR)



Evaluation on Test Set (Best SVR):

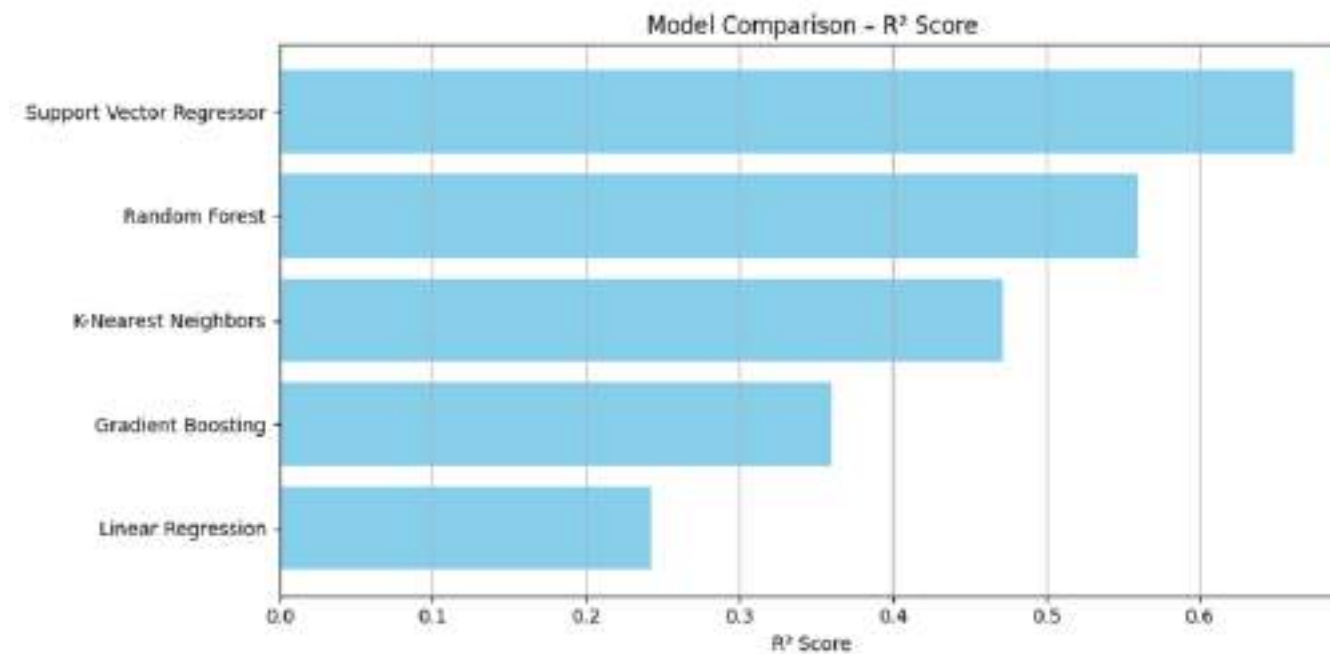
R^2 Score: 0.6984

MAE : 162.97

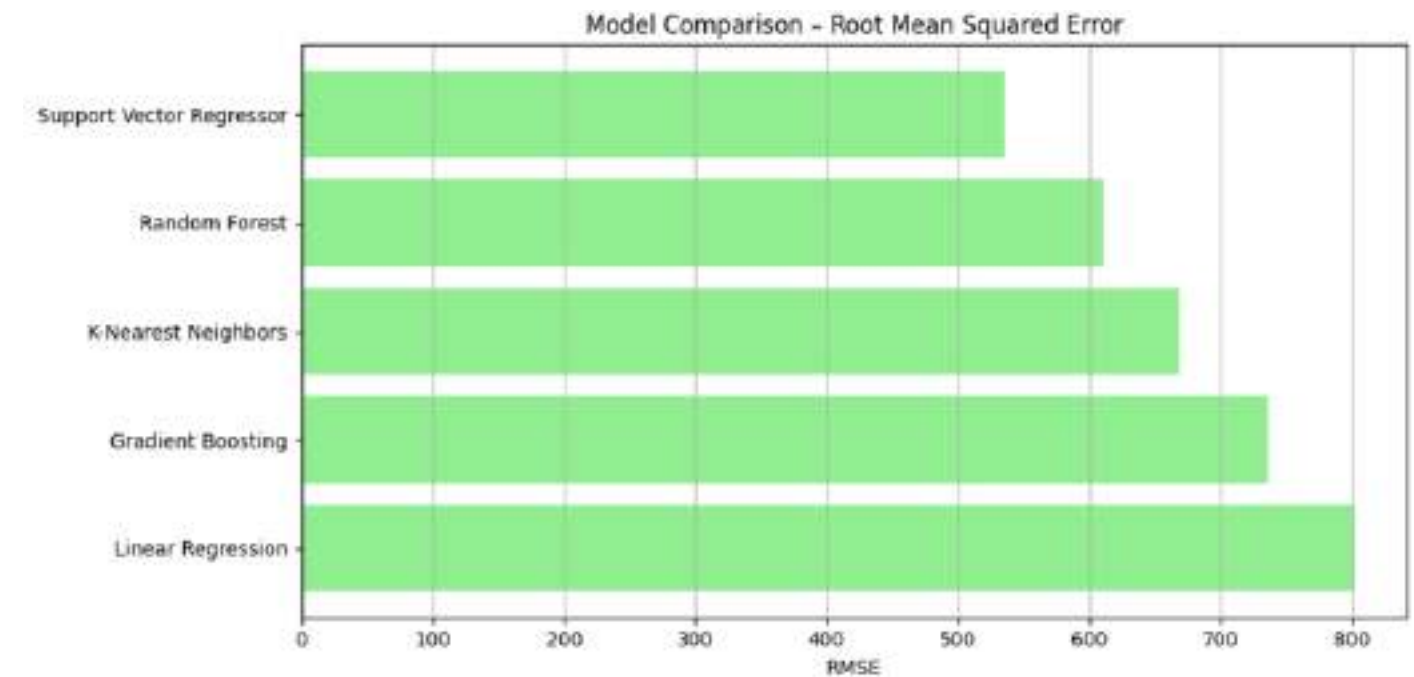
RMSE : 505.93

MODEL EVALUATION & COMPARISON

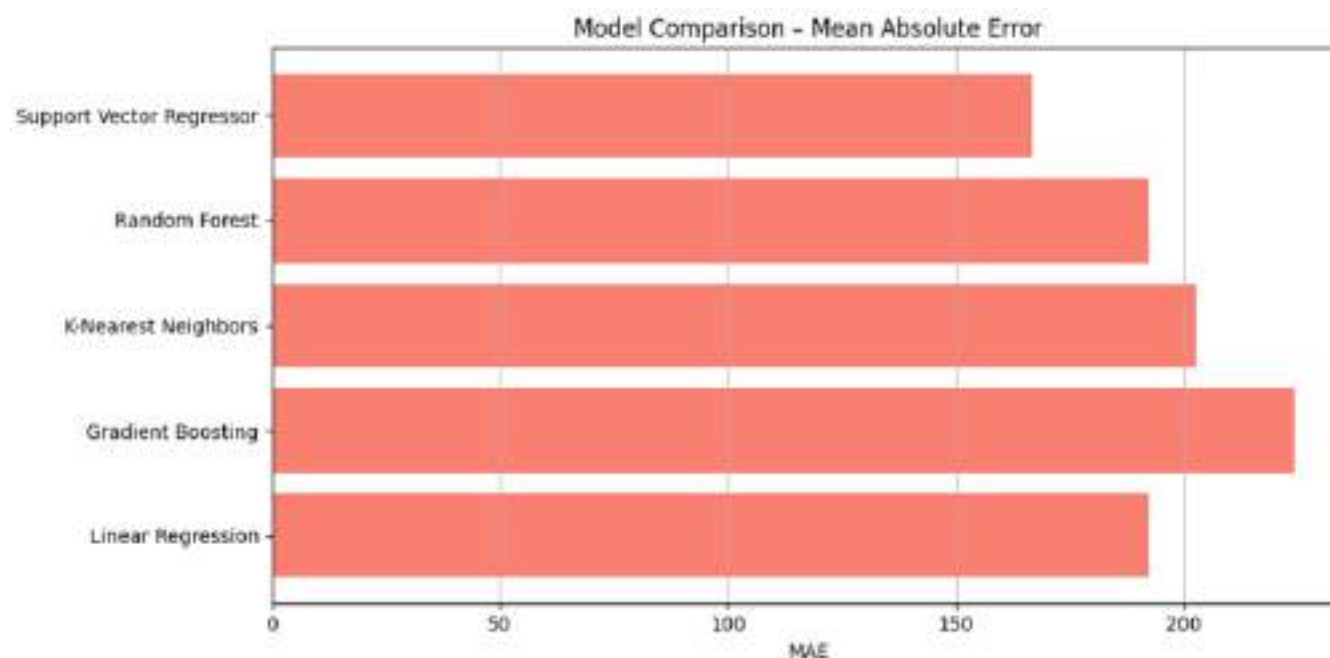
R^2 Score



Root
Mean
Squared
Error
(RMSE)



Mean
Absolute
Error
(MAE)



Based on these 3 metrics, we can confidently say that the Support Vector Regression (SVR) model is the best model.

OUR MODEL VS AUTOSKLEARN

Feature	Our model	auto-sklearn
Model selection and optimisation	Manually selects algorithms and tunes hyperparameter	Automated model selection, preprocessing and hyperparameter tuning
Overall performance	Fully dependent on manual tuning	Offers higher accuracy by leveraging automated ensemble methods

From here, we can confidently state auto-sklearn is likely to outperform our selected model across all metrics, thanks to its automated optimisation over a broad search space and use of ensemble methods.

CONCLUSION

- **Support Vector Regression (SVR)** emerged as the model with best performance, with an R^2 score of 0.7414 after hyperparameter tuning using GridSearchCV.
- **Planted area** is the most influential feature.
- **Temperature-humidity index (THI)** improved model predictions.
- **District-specific insights** (like long bean yielding highest in Batu Pahat) support the hypothesis that environmental and regional factors are crucial in crop production.



THANK YOU